

## Problem Sets Before Test 1

Only turn in problems that are **not** bracketed. Bracketed problems are additional problems you can look at. Round brackets indicate problems that may help you with problems that are assigned; square brackets are additional problems on material that you should know, but you are not required to write up solutions; curly brackets are truly optional and may contain extra nuggets that you will not be required to know but may be interested in.

Additional assignments will be filled in over time.

| notation    | meaning                                                                       |
|-------------|-------------------------------------------------------------------------------|
| unbracketed | assigned problem – turn these in for grading                                  |
| ()          | helper/warm-up problem                                                        |
| []          | additional problems (you are responsible for content, but don't turn them in) |
| { }         | covers optional material                                                      |

| Due      | Task                 | Source     | Problems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|----------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| never    | <a href="#">WW00</a> | –          | Intro to WebWork                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Wed 1/11 | miscellaneous        | online     | <p>1 Visit the class web page at <a href="https://calvin-m252-s23.netlify.app/">https://calvin-m252-s23.netlify.app/</a></p> <p>2 Fill out this <a href="#">pre-course survey</a></p> <p>3 Check that you have access to important course-related things:</p> <ul style="list-style-type: none"> <li>Check that you can login to <a href="#">WebWork</a> using your usual Calvin login and password.</li> <li>Check that you can access <a href="https://gradescope.com">https://gradescope.com</a>. You should also have received an email invite.</li> <li>Check that you received an email from me addressed to the class.</li> </ul> <p>If any of these things are not working for you, let me know.</p> |
|          | PS 1                 | Rosen 10.1 | Submit the unbracketed problems via Gradescope<br>(1) (3,5,7,9) types of graph<br>4 types of graph 8 types of graph 13 intersection graph 18 Fred 33 precedence graph                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Fri 1/13 | <a href="#">WW01</a> | Rosen 10.2 | Do these problems in WebWork.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mon 1/16 | <a href="#">WW02</a> | Rosen 10.3 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Wed 1/19 | PS 2                 | Rosen 10.3 | 9ace special graphs [10–11] adjacency matrix 12 adjacency matrix<br>[35–43 odd] isomorphic? 36 isomorphic? 38 isomorphic? 40 isomorphic?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Fri 1/21 | PS 3                 | Rosen 10.4 | 2 paths 6 components 12 strong/weak connectivity 14 components<br>22 isomorphic? 64 farmer puzzle 66 water puzzle<br>[1] paths [3–5] [11] strong/weak connectivity [21, 23] isomorphic?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |